

Name – Mukund Raj

Roll No. - 23b4245

Week 1 Assignment

1. Five Interesting Concepts I Learned from the Computer Vision Lectures

1. Human Vision Inspired Computer Vision

One of the most interesting things I learned is that computer vision is heavily inspired by human vision. Our eyes evolved over millions of years, starting from simple light-sensitive spots to the complex eyes humans have today. Computer vision researchers study how humans see and understand the world and try to build systems that can do similar tasks.

2. Different Levels of Computer Vision

I learned that computer vision can be divided into three levels:

- Low-level vision deals with pixels and basic image operations such as resizing, filtering, and edge detection.
- Mid-level vision focuses on understanding relationships between images, such as creating panoramas and reconstructing 3D scenes.
- High-level vision is about understanding the meaning of images, such as identifying objects, recognizing faces, and segmenting different parts of an image.

This helped me understand how computer vision systems gradually move from raw pixels to meaningful information.

3. Color Spaces Are Important

Before watching the lectures, I thought images were only represented using RGB (Red, Green, Blue). I learned that there are many other color spaces such as HSV and LAB, each designed for specific tasks. Different color spaces make it easier for computers to analyze colors, detect objects, and process images efficiently.

4. Image Resizing Is More Complex Than It Looks

I learned that resizing an image is not as simple as stretching it. Computers use methods such as nearest-neighbor interpolation and bilinear interpolation to estimate new pixel values. Different techniques produce different image qualities, and choosing the right method is important in image processing applications.

5. Deep Learning Has Revolutionized Computer Vision

The lectures explained how recent advances in neural networks have significantly improved computer vision. Tasks that were once very difficult, such as object detection, image classification, and segmentation, can now be performed with high accuracy using deep learning models. This showed me why modern AI systems are so powerful in understanding images.

2. Research on Healthcare Reports and Medical Imaging Reports

Types of Healthcare Reports

Healthcare organizations use many different types of reports to record patient information and assist in diagnosis and treatment. Some common healthcare reports are:

1. Laboratory Reports
2. Radiology Reports
3. Pathology Reports
4. Discharge Summaries
5. Surgical Reports
6. Prescription Records
7. Clinical Notes
8. Medical Imaging Reports

Each report serves a different purpose and helps healthcare professionals make informed decisions.

Medical Imaging Reports

Medical imaging reports are documents prepared by radiologists after examining medical images. These reports summarize the findings observed in scans and help doctors diagnose diseases and plan treatments.

Medical imaging is widely used because it allows doctors to see inside the human body without performing surgery.

Common Types of Medical Imaging

- X-Ray
- CT Scan (Computed Tomography)
- MRI (Magnetic Resonance Imaging)
- Ultrasound
- PET Scan
- Mammography

Structure of a Medical Imaging Report

A typical medical imaging report contains:

1. Patient Information
2. Examination Details
3. Clinical History
4. Findings
5. Impression or Conclusion
6. Recommendations

The findings section describes what the radiologist observes in the image, while the impression section provides the final interpretation.

Importance of Medical Imaging Reports

Medical imaging reports are important because they:

- Help doctors diagnose diseases accurately.
- Support treatment planning and monitoring.
- Improve communication between healthcare professionals.
- Maintain a permanent medical record for future reference.